

Software Engineering
Phase 1 – Software Requirements Specification

Team-E16 — Traffic Management System (Simulation)

Section E

Team Members:-

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1. Problem Statement, Analysis & Feasibility

1.1 Problem Statement

Urban road networks frequently suffer from congestion at intersections due to static, fixed-time traffic signals that do not adapt to real-time vehicle load. This leads to increased waiting times, fuel wastage, and delayed movement of emergency vehicles. There is a need for a system that simulates and manages traffic flow at intersections using efficient data structures, allowing operators to monitor congestion, adjust signal timing, and prioritise emergency vehicles.

The Traffic Management System (Simulation) is a C/C++ application that models vehicle arrival and departure at intersections using queues, dynamically manages signal timing, simulates traffic flow, and generates congestion reports to aid decision-making.

1.2 Feasibility Analysis

Technical Feasibility:

- The system is implementable in C/C++ using core data structures — queues for vehicle lanes, priority queues for emergency handling, and arrays/structs for intersection state.
- No external hardware is required; the simulation runs as a console/desktop application, making it feasible within an academic environment.

Economic Feasibility:

- Zero licensing cost — built with open-source GCC/G++ toolchains, keeping the project financially viable for an academic setting.

Operational Feasibility:

- The interface is simple enough for a traffic operator to monitor signals and adjust timing. Reports are generated in a readable format for analysis.

Schedule Feasibility:

- The scope is modular (simulation engine, signal controller, reporting) and can be delivered iteratively across Phases 2–5.

2. Software Requirements Specification (SRS)

2.1 Purpose

This document specifies the functional and non-functional requirements for the Traffic Management System (Simulation), intended to model and control traffic flow at intersections.

2.2 Scope

The system simulates vehicle movement through intersections using queue-based data structures, controls traffic signal timing, prioritises emergency vehicles, monitors congestion, and produces traffic reports. It is a single-machine simulation and does not interface with live traffic hardware.

2.3 Requirements List

ID	Requirement
R1	The system shall simulate vehicle arrival and departure at each lane of an intersection using queues.
R2	The system shall allow the operator to control the timing (green/red duration) of traffic signals.
R3	The system shall monitor and display the real-time congestion status of each lane/intersection.
R4	The system shall detect and prioritise emergency vehicles over normal traffic.
R5	The system shall allow configuration of intersection parameters (number of lanes, arrival rate, signal cycle).
R6	The system shall generate traffic reports summarising throughput, average wait time, and congestion.
R7	The system administrator shall manage user accounts and system-wide configuration.
R8	The system shall respond to a user command within an acceptable time and remain stable during simulation.
R9	The system shall validate all inputs and handle invalid configuration gracefully.
R10	The system shall protect operator/admin functions behind authenticated access.

3. Requirement Categorization (Functional & Non-Functional)

3.1 Functional Requirements (FR)

FR ID	Functional Requirement
FR1	Simulate traffic flow at intersections using queue data structures.
FR2	Control and adjust traffic signal timing for each lane.
FR3	Monitor and display congestion status per lane/intersection.
FR4	Detect emergency vehicles and grant them signal priority.
FR5	Configure intersection parameters (lanes, arrival rate, signal cycle).
FR6	Generate and export traffic reports (throughput, wait time, congestion).
FR7	Manage user accounts (create, authenticate operator/admin).

3.2 Non-Functional / Validation Requirements (NFR / VFR)

NFR ID	Non-Functional Requirement	Category
NFR1	The system shall respond to operator commands within 1 second during simulation.	Performance
NFR2	The simulation shall remain stable for extended runs without crashing.	Reliability
NFR3	Operator and admin functions shall require authentication.	Security
NFR4	The console interface shall be clear and easy for an operator to use.	Usability
NFR5	The code shall be modular so components can be maintained independently.	Maintainability
NFR6	Invalid inputs shall be rejected with meaningful error messages.	Robustness

4. Requirements Traceability Matrix (RTM)

Req ID	Description	Maps To (FR/NFR)	Use Case
R1	Simulate vehicle arrival/departure	FR1	Simulate Traffic Flow

Req ID	Description	Maps To (FR/NFR)	Use Case
R2	Control signal timing	FR2	Control Signal Timing
R3	Monitor congestion status	FR3	View Congestion Status
R4	Emergency vehicle priority	FR4	Handle Emergency Priority
R5	Configure intersection params	FR5	Configure Parameters
R6	Generate traffic reports	FR6	Generate Traffic Reports
R7	Manage user accounts	FR7	Manage User Accounts
R8	Response time & stability	NFR1, NFR2	All
R9	Input validation	NFR6	Configure Parameters
R10	Authenticated access	NFR3	Manage User Accounts

5. Actors

#	Actor	Description
1	Traffic Operator	Monitors signals, controls timing, runs and observes the simulation, handles emergency priority, and generates reports.
2	System Administrator	Manages user accounts, configures system-wide parameters, and manages intersections.
3	Commuter	Views congestion status and traffic reports (read-only end user).

Number of actors: 3

6. Use Cases Grouped by Actor

6.1 Traffic Operator

- Monitor Traffic Signals
- Control Signal Timing
- Simulate Traffic Flow (Queues)
- Manage Intersections
- Handle Emergency Vehicle Priority
- Generate Traffic Reports

6.2 System Administrator

- Manage User Accounts
- Configure System Parameters
- Manage Intersections

6.3 Commuter

- View Congestion Status
- Generate Traffic Reports (view)

7. Use Case Diagram

